$$\frac{\partial}{\partial t}\rho_{\mathcal{S}}(X,X',t)=\left[-\frac{\mathrm{i}}{2M}\left(\frac{\partial^2}{\partial X'^2}-\frac{\partial^2}{\partial X^2}\right)-\frac{\mathrm{i}}{2}M\left(\Omega^2-2\gamma_0\Lambda\right)\left(X^2-X'^2\right)\right.$$
$$\left.-\gamma_0\left(\frac{X-X'}{\lambda_{\mathrm{dB}}}\right)^2\right]\rho_{\mathcal{S}}(X,X',t). \quad (5.63)$$